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High Viscosity Friction Reducer Enhanced Oil Production in Reservoirs with Moderate to High Permeability

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Abstract

Fracturing fluids have evolved through time due to various challenges present in the oil and gas industry. These challenges make it difficult to find one fluid that works for all type of formations. Some of the most important and determining parameters that dictates the type and properties of the fracturing fluid that will be used are the formation permeability, native fluids and desired geometry. Type of leak-off for fracture geometry generation, behavior of net pressure and viscosity for proppant transport are three major parameters to consider in the design of the fluid. This will ensure that the generated fracture provides high conductivity to maximize production or injection rate, while formation damage is minimized.

Leak-off and viscosity requirements from a fracturing fluid vary for low and high permeability formations. The design of a fluid that meets the characteristics of both low and high permeability formation is challenging.

Success applications of the next generation of high viscosity friction reducer technology as main fluid in propped Hydraulic fracturing jobs to improve oil production in mature fields located in Colombia is presented. It has been a major challenge because these reservoirs have permeabilities from 90 to 1500 milidarcies, very low reservoir pressure gradients of 0.23 to 0.25 psi/ft and their producer wells with BSW higher than 90%. The lithological logs show reservoirs with clean sandstones in the lower zone meanwhile in the upper zone shows intercalations of sand and shale.

The next-generation HVFR technology was designed and proposed specifically for these projects. This HVFR was developed with the characteristics of a crosslinked fracturing fluid showing a viscosity in the range of 200-400 cP and can transport proppant concentrations up to 5ppa. Prior to field application, core-flooding tests showed a retained permeability to oil higher than 90% meanwhile compatibility of the system with the native fluids as was verified.

The HVFR fluid was pumped in four wells (three producer and one injector wells). These wells were successfully completed, pumping proppant concentrations up to 6 ppg. Total fluid breakthrough (viscosities less than 10 cP) and optimal fracture geometries for production were obtained decreasing the average of well-testing along flowback stage. In all the producer wells their oil production improved, and their perdurability was observed by at least 6 months.

Introduction

In the last few decades, propped hydraulic fracturing has been used to access to oil and gas trapped in low permeability formations or to bypass the skin damage in the nearwellbore and enhance productivity/ injectivity indexes.

Geometry of fracture depends on type of fluid and its properties. Viscosity, fluid crosslinking, proppant distribution and suspension, salinity tolerance, nanoparticles tolerance and temperature stability have been optimized to mitigate the negative effects in the fracture conductivity. Also, static bottomhole temperature, petrophysical properties, reservoir pressure and the mechanism of leak-off are some properties that determine the fracturing fluid design to each scenario.

In the east (scenario 1) and south (scenario 2) of Colombia, moderate to high permeability sandstones (from 150 to more than 1500 milidarcies) demands robust fracturing fluids to create fractures more conductive than these reservoirs and transport high proppant concentrations (over up 6 pounds per gallon, ppa.). The selected fluid must be ensured to control the rate of leak-off into the rock and create the proper fracture geometry by the optimization of the proppant quantity. To achieve this, polymeric loading was increased by more than 35 pounds per thousand gallon (ppg) to achieve the right stability against leak-off (rheology test). This higher loading affects the fracture conductivity (residues inside the fracture) as well as drives an increase in surface pressure due to friction along the frac string that will increase the required hydraulic horsepower (HHP) to execute the designed frac job.

The objective here was to develop the next generation of HVFR fluid: cleaner, more stable, free of guar and its derivatives and optimize the number of additives. For instance: eliminate the use of activators (that are conditional to changes in temperature and injection water composition). This fluid mitigates the generation of residue into the fracture, transmits an effective pressure inside the fracture pack and reduces the friction along the frac string during the pumping of hydraulic fracturing Job.

Statement of theory and definitions

High viscosity friction reducer (HVFR) fluids were originally used in propped hydraulic fracturing jobs for low and very low permeability reservoirs like shale and tight formations (conventional and unconventional reservoirs of oil or/and gas). Understanding some of its outstanding features correspond to the low generation of residue inside the fracture, high proppant transport efficiency and reducing up to 85% of friction losses (it means more effective pressure inside the formation).

Using the next generation of HVFR as fracturing fluid in the moderate to high reservoirs in the south and east of Colombia some advantages over conventional fluids like guar gum and its derivatives have been observed:

a. Reduction of service and material costs. Less quantity of additives and less volume per each additive. Less equipment in the frac fleet, for instance a hydration unit is not required.
b. Compatibility with almost all the oxidizing breakers available in the local and global markets. For instance: delayed oxidizers, encapsulated oxidizers, and conventional oxidizers.
c. Less residual polymer after breaking down of fracturing fluid. Demonstrated by core-flooding tests.
d. High proppant transport capacity higher viscosity at low shear rates than conventional fluids.
e. Flexible designs with the use of industrial water.

Rheology and Proppant Transport Capacity

Predicting the behavior of fracturing fluids under different scenarios of shear rates allow higher control during the fracturing operation. Geometry of fracture, efficiency of fluid, leak-off rate and spurt loss are addressed directly by characteristics of fracturing fluid. At the same time efficiency of fluid is affected by reservoir pressure and permeability of reservoir. Higher permeability has lower fluid efficiency and requires more robust fracturing fluids. Lower reservoir pressure has lower fluid efficiency and more viscous fluids are required to transport the proppant.

"Considering that the shear rate is low inside the fracture, and ranges between 0.1 and 10 s-1, we can observe the viscosity values are clearly higher in a HVFR fluid than a linear gel (Figure 1), giving it a substantial advantage in this sense for proppant transport. At higher shear rates, similar viscosities are observed in both fluids" (Azcurra, Marcos; Bach, Roberto,2022). Based on this, increased slurry rates are not required to transport the proppant properly when HVFR is used as fracturing fluid.

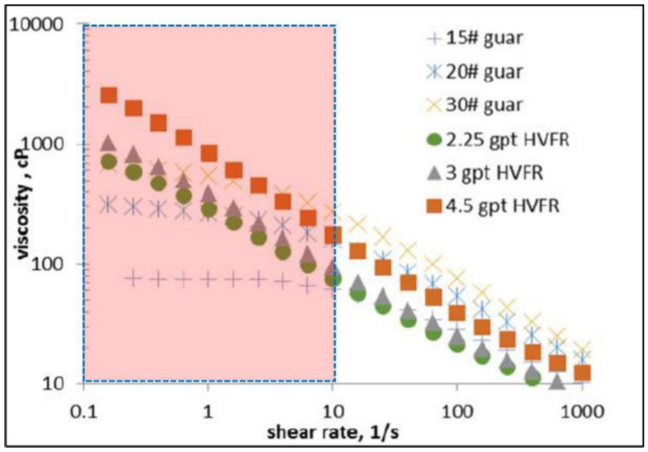


Figure 1—Viscosity as function of the shear rate, for HVFR and Guar based fluids. At low shear rates, HVFR develops higher viscosity than guar-based fluid, giving greater transport capacity. Taken from SPE-189841-MS.

High tolerance for salinity, hardness and dissolved solids

Guar and HVFR fluids are sensitive to high content of dissolved solids like sulfates, iron and carbonates (hardness). Since injection water in scenario 1 (east of Colombia) and scenario 2 (south of Colombia) had high total dissolved solids (TDS), additives were included that helped with fast hydration and developing the required viscosity for optimal proppant transport. Table 1 shows the injection water composition used in the projects. Additional lab tests were performed ensuring compatibility between HVFR and reservoir fluids.

Table 1—(a) Characteristics of Injection Water. Scenario 1.

Parameter	Value	Dimension
Turbidity	11.5	NTU
Density	1.0025	g/mL
pH Range	8.0 – 8.1	dimensionless
Chlorides	56	mg/L
Total Hardness	13.8	mg/L
Total Alkalinity	91	mg/L
Sulfates	21	mg/L
Iron	2.04	mg/L

Table 1—(b) Characteristics of Injection Water. Scenario 2.

Parameter	Value	Dimension
Turbidity	4.4	NTU
Density	1.015	g/mL
pH Range	6.7 – 7.0	dimensionless
Chlorides	6200	mg/L
Total Hardness	1740	mg/L
Total Alkalinity	1500	mg/L
Sulfates	0	mg/L
Iron	0.99	mg/L

Retained Permeability

Core flow testing results refer to the final or retained permeability, which is measured after each fluid phase was displaced into the rock after treatment and compared to initial permeability. Values higher than 80% are good enough results in reservoirs with moderate to high Permeability.

Service and material costs

In the last few years raw materials, logistic, shipment and storage processes have affected the final cost of additives and services used in the fracturing operations. Reduction of equipment in the fracturing fleet and quantity of additives are achieved by the optimization or elimination of stabilizers, breakers and crosslinkers in the HVFR fluids.

Settling Velocity and Suspension of Proppant

The settling rate of the proppant is directly proportional to the difference in density between the fracturing fluid and proppant and inversely proportional to the fluid viscosity (Speight et al. 2016). Under atmospheric conditions settling velocity of linear gel (guar gum with 25 lb/Mgal) is close to 0.06 mm/s (millimeters per second). Achieving similar results using other fluids like HVFR are recommended to avoid bridging inside the fracture and premature screen outs. Also, the higher the proppant suspension capacity the higher the travel of proppant into the fracture and the higher the conductivity along the fracture.

Presentation of data and Results

Processes related to the lab testing and field applications are discussed below in support of the data and results.

Laboratory Tests

Prior to the field application lab tests like compatibility, settling velocity, rheology and core-flooding tests were performed to ensure an HVFR developed under the requirements for both projects. For the scenario 2, a thinner HVFR was tested: less initial viscosity and quicker breakdown time. On the other hand, for the eastern reservoirs (scenario 1) a robust HVFR was used: initial viscosities close to 400 cp and breakdown times higher than 90 minutes.

Compatibility Test. Screening, compatibility, clean up, interfacial tension and wettability were ensured between HVFR and native fluids like water and oil formation. In summary, clean detergency and interfacial tensions lower than 1.0 mN/m were observed along the tests. No emulsions or incompatibilities were observed.

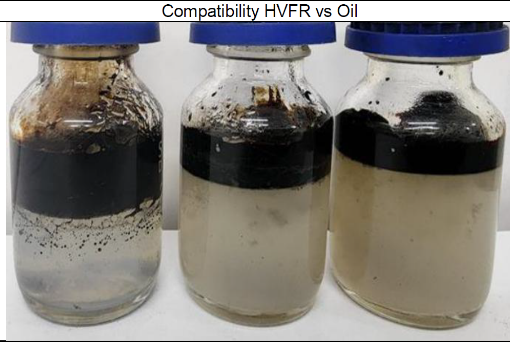
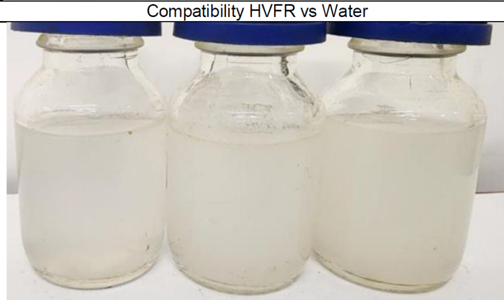
<table>
<tr><th rowspan="2">Compatibility Test</th><th colspan="6">Separation of phases at time (minutes)</th><th rowspan="2">Comments</th></tr>
<tr><th>5</th><th>10</th><th>15</th><th>20</th><th>30</th><th>60</th></tr>
<tr><td>Vs reservoir Oil</td><td>60</td><td>95</td><td>99</td><td>100</td><td>100</td><td>100</td><td>Two phases</td></tr>
<tr><td>Vs reservoir Water</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>One phase</td></tr>
<tr><td colspan="3">Wettability Test</td><td colspan="5">Wettability to water</td></tr>
<tr><td colspan="3">Detergency</td><td colspan="5">Good</td></tr>
<tr><td colspan="8">Records (Pictures)</td></tr>
<tr><td colspan="3">Compatibility HVFR vs Oil</td><td colspan="3">Filtrate through a 325 mesh</td><td colspan="2">Detergency</td></tr>
<tr><td colspan="3"></td><td colspan="3">Reservoir oil is very viscous and does not pass through the mesh</td><td colspan="2" rowspan="3"></td></tr>
<tr><td colspan="3">Compatibility HVFR vs Water</td><td colspan="3">Filtrate through a 325 mesh</td></tr>
<tr><td colspan="3"></td><td colspan="3"></td></tr>
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Figure 2a—Compatibility Test. Scenario 1

<table>
<tr><th rowspan="2">Compatibility Test</th><th colspan="6">Separation of phases at time (minutes)</th><th rowspan="2">Comments</th></tr>
<tr><th>5</th><th>10</th><th>15</th><th>20</th><th>30</th><th>60</th></tr>
<tr><td>Vs reservoir Oil</td><td>90</td><td>100</td><td>100</td><td>100</td><td>100</td><td>100</td><td>Two phases</td></tr>
<tr><td>Vs reservoir Water</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>One phase</td></tr>
<tr><td colspan="3">Wettability Test</td><td colspan="5">Wettability to water</td></tr>
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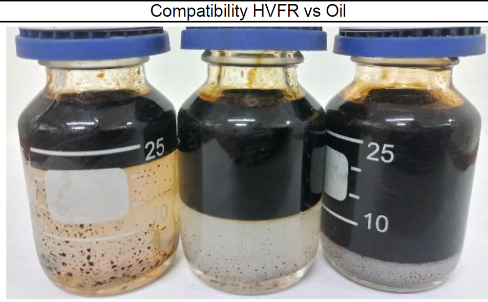
<tr><td colspan="3">Compatibility HVFR vs Oil</td><td colspan="3">Filtrate through a 325 mesh</td><td colspan="2">Detergency</td></tr>
<tr><td colspan="3"></td><td colspan="3"></td><td colspan="2" rowspan="3"></td></tr>
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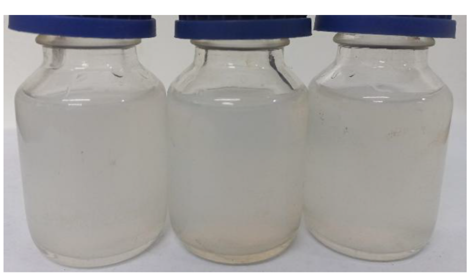
<tr><td colspan="3"></td><td colspan="3"></td></tr>
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Figure 2.b—Compatibility Test. Scenario 2.

Rheology Test. To perform these tests, bottomhole temperature and 100 s-1 as shear rate were used following API standard ***(API RP 39).*** Assuming a pumping schedule of 40 minutes for the reservoir with high permeability (scenario 1), an HVFR fluid was developed with the following features:

a. Static Bottomhole Temperature: 185 Fahrenheit
b. Initial Viscosity: from 300 to 400 cP
c. Vortex time: 30 seconds
d. Breakdown time under dynamic conditions: 90 minutes
e. Breakdown time under static conditions: less than 240 minutes

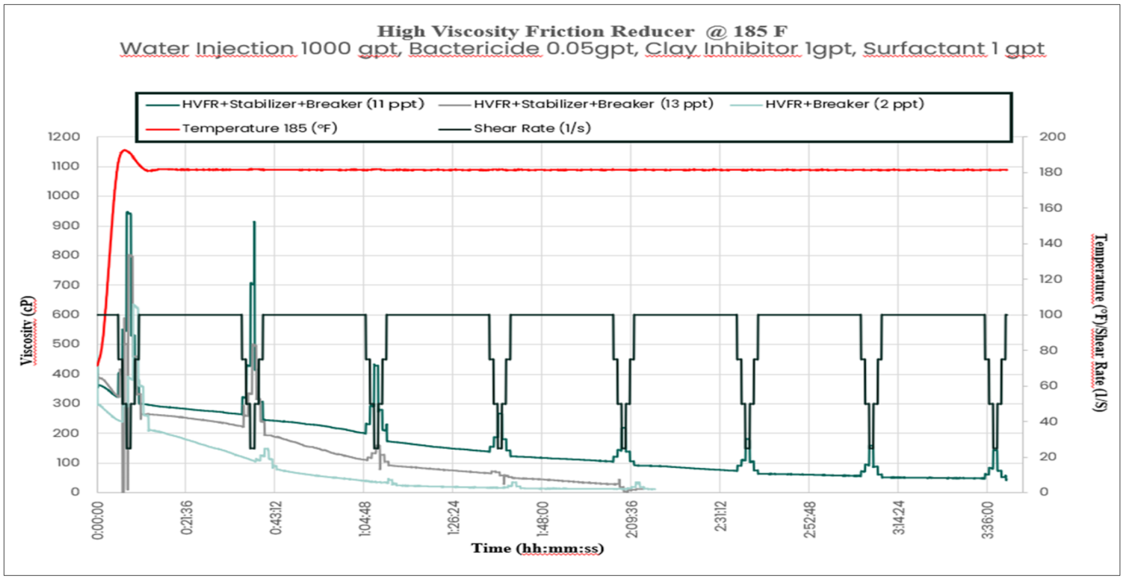

Figure 3.a—Rheology Curve. Scenario 1

For scenario 2 (moderate permeability) the pumping schedule of the Fracturing Job was only 20 minutes. Based on that timing, an HVFR fluid was developed with the following features:

a. Static Bottomhole Temperature: 125 Fahrenheit
b. Initial Viscosity: from 250 to 300 cP
c. Vortex time: 50 - 65 seconds
d. Breakdown time under dynamic conditions: 60 minutes
e. Breakdown time under static conditions: less than 180 minutes

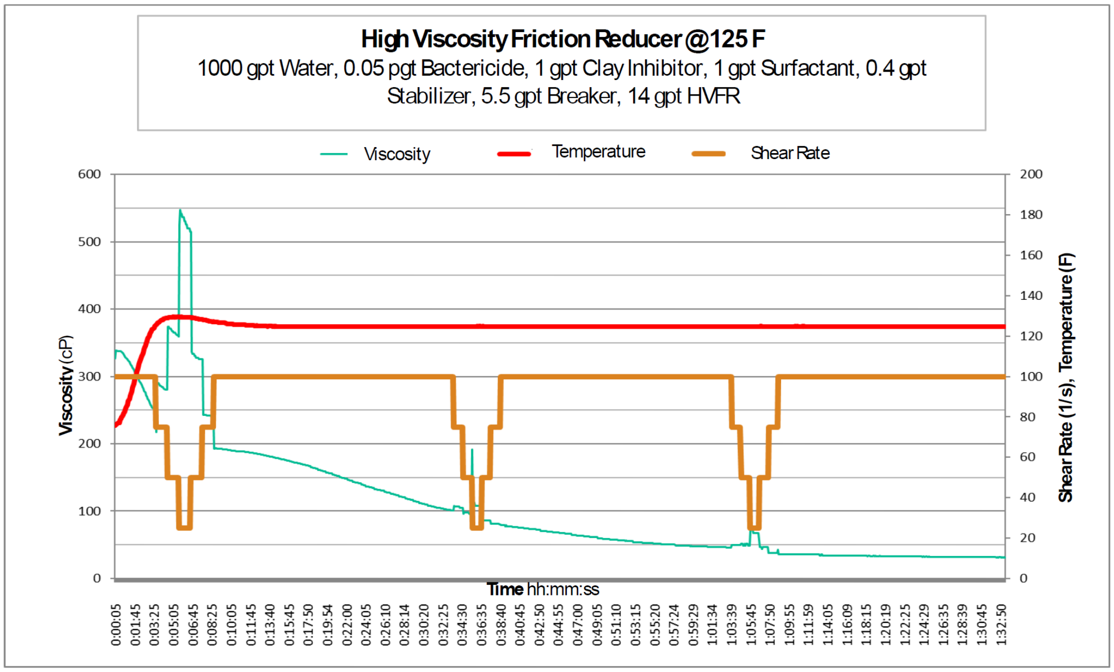

Figure 3.b—Rheology Curve. Scenario 2

Settling Velocity Test. Since the density of HVFR (1.0 g/cm3) is less than half of density ceramic proppant (2.6 g/cm3), the viscosity of HVFR plays a major role to keep suspended the proppant along the time. Under static conditions, ceramic proppant was suspended at least 90 minutes by HVFR fluid. This test simulated the time that fracture would be opened during frac operation in the reservoir located in the east of Colombia. Regarding the operation of south of Colombia (scenario 2), proppant suspended for 60 minutes would be enough for the execution of the frac job operation.

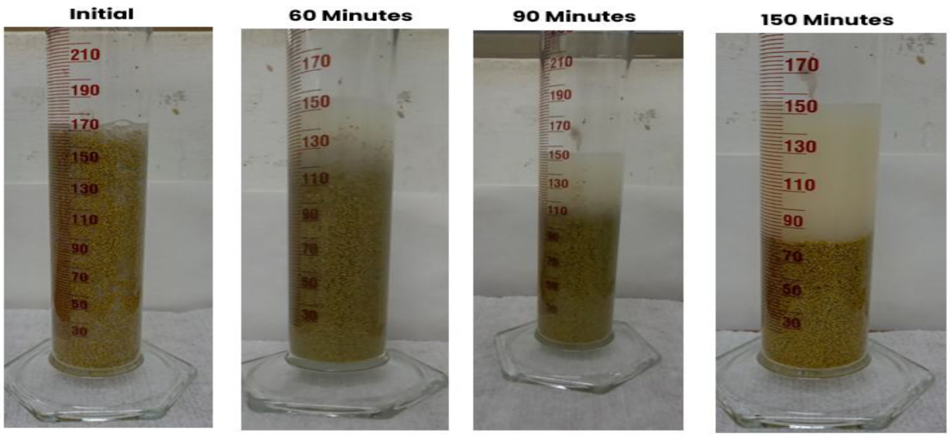

Figure 3—Settling Velocity Test. 20/40 Ceramic Proppant

Core-flooding Test. Core-flooding test evaluated the dynamic efficiency of the next generation of HVFR, by injecting at least ten pore volumes of broken fluid, which showed that in general the fracturing fluid yielded good performance by reaching oil permeability recovery greater than 90% with respect to the permeability baseline.

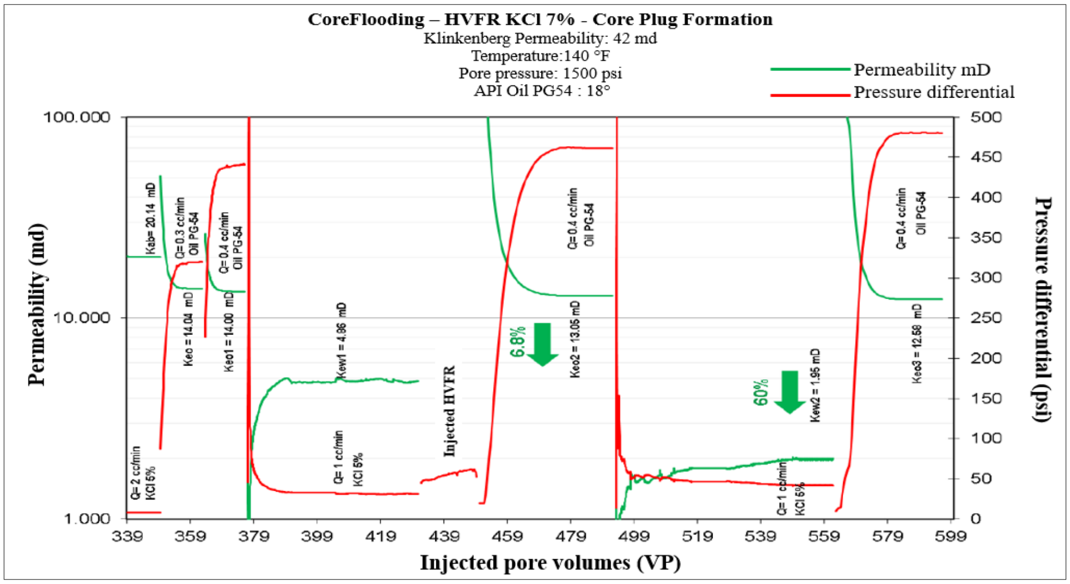

Figure 4—Retained Permeability Test. Scenario 2

Table 2—Efficiency Results of HVFR fluid in Core-flooding test

Efficiency Results of HVFR fluid in Core-flooding test					
Initial Kew (Baseline)	Final Kew	% Retained of Kew	Initial Keo	Final Keo	% Retained of Keo
4.96	1.95	39.9	14.04	12.68	90.3

Field Applications. Based on lab test results, two different reservoirs were selected to perform propped hydraulic fracturing jobs using the HVFR. The first reservoir (east of Colombia) is divided in two different sandstones. Since its pressure reservoir is about 1650 psi/ft (measured at true vertical depth of 6600 feet and corresponds to 0.25 psi/ft), this is considered a depleted reservoir. Its lower sand is a clean reservoir with homogenous petrophysical (permeabilities from 500 to 4000 mD) and geomechanical properties along all the intervals. The upper layer is made up of intercalations of dirty and clean sands making it a heterogenous reservoir with permeabilities between 35 to 1600 mD. In both lower and upper sands, the mobility to oil is lower than to water.

The second reservoir is composed by two packs of intercalations of dirty and clean sands, also. The number of layers for every sand depends on how the sedimentary environment was and where is located inside the field. Porosity can vary from 9 to 17%, while permeability is between 50 to 950 mD. At true vertical depth of 2800 ft, its reservoir pressure is closed to 650 psi, corresponding to 0.28 psi/ft.

In both scenarios, the baseline of the fracturing designs corresponded to those past fracturing jobs where guar gelling, and derivatives were used. A volume of PAD lightly higher was considered: 40% of PAD in the first scenario and 25% of PAD for the other case. The slurry rate and total mass of proppant didn't change, and salinity of base fluid was the same (1 gal/Mgal of clay inhibitor was used). For the first reservoir (minimum horizontal stress was 3300 psi or 0.5 psi/ft), 20/40 pre-coated ceramic was selected as main proppant. Meanwhile for the fracture in the second reservoir (minimum horizontal stress was 1560 psi or 0.57 psi/ft), 16/30 natural sand was the proppant selected.

Details of Scenario 1

to calibrate the fracture model a step-down test was performed as injection test. A total volume of 580 Bbl of HVFR were pumped at maximum clean rate of 35 bpm. The maximum surface pressure and bottomhole pressure were 3550 psi and 5440 psi, respectively. Table 3 shows the summary of pressure declination and step-down test analysis. This data was used to perform the match of surface and bottomhole pressures (figure

5) and update the fracturing model using the fracturing software. The HVFR showed 55% of fluid efficiency and 130psi/1000ft as tubing frictions at 35 bpm. Finally, field operation (figure 6) was executed at slurry rate of 35 bpm, and 580 Bbl of HVFR and 60K pounds of 20/40 pre-coated ceramic were pumped in the frac job. The observed net pressure showed a mode III, suggesting a tip screen-out packing of the fracture. Usually, fracture geometries (figure 7) trends to height and width growths supported by lower contrasts in the geomechanical parameters and moderate static young's modulus and evidenced by temperature logs. Table 4 shows the main parameters of the fracture geometry estimated in the scenario 1.

Table 3—Analysis of Step Down Test. Scenario 1.

Parameter	Value	Dimension
Measured Depth	7460	feet
True Vertical Depth	6969	feet
Surface Pre-ISIP (Instantaneous Shut-In Pressure)	3500	Psi
Surface ISIP	1929	Psi
Bottomhole ISIP	4948	Psi
Fracture Gradient	0.71	Psi/feet
Closure Stress	3756	Psi
Closure Stress Gradient	0.54	Psi/feet
Closure Time	11.1	Minutes
Net Pressure	1191	Psi
Fluid Efficiency	55	%
Total Friction at 30 bpm	1571	Psi
Frac String friction at 30 bpm	980	Psi
Entry Hole friction at 30 bpm	591	Psi
Friction at perforations at 30 bpm	319	Psi
Nearwellbore friction at 30 bpm	272	Psi

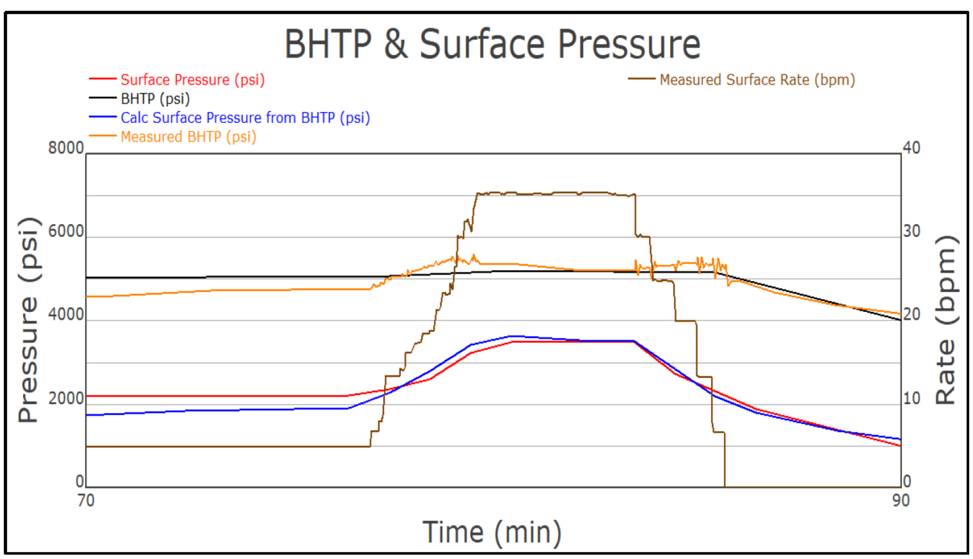

Figure 5—Match of Pressures. Injection Test. Scenario 1.

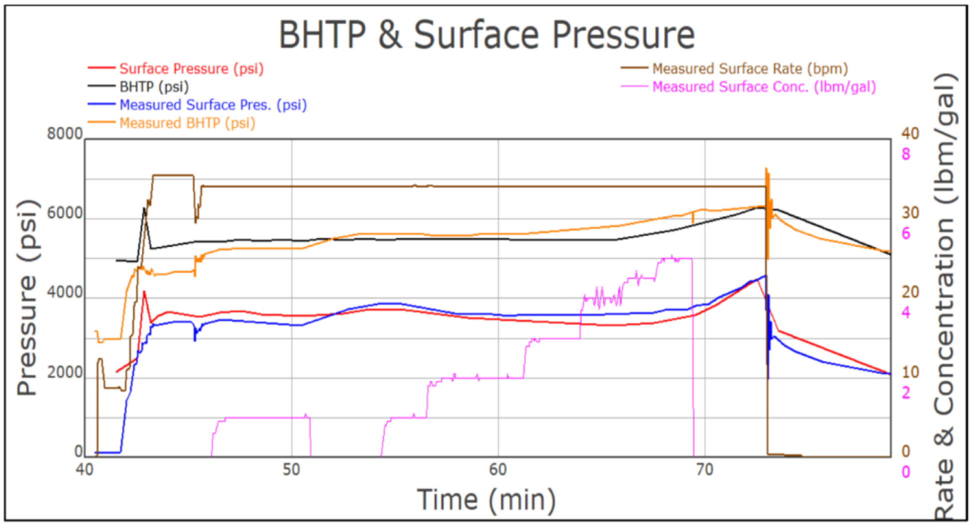

Figure 6—Parameters of Fracturing Job. Application of HVFR in the field. Scenario 1.

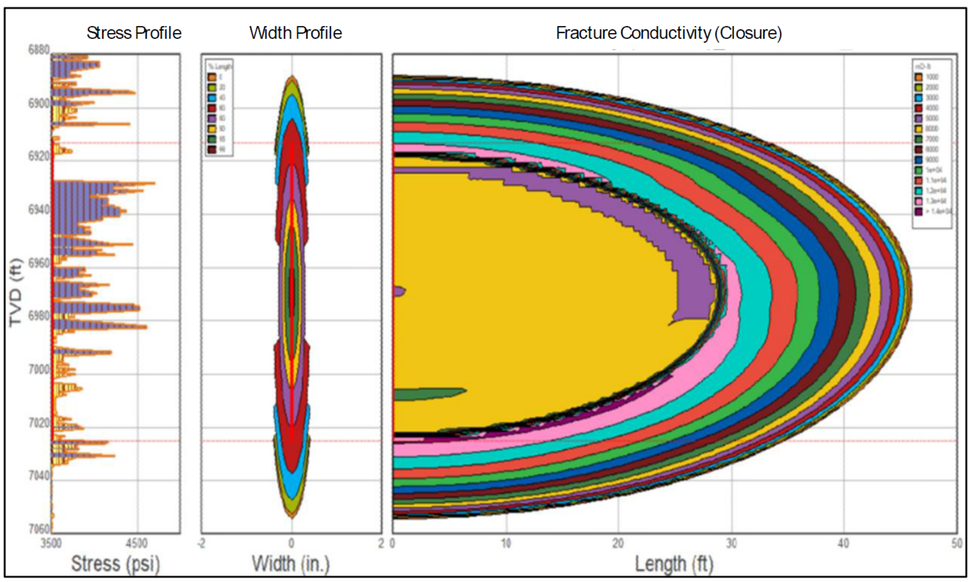

Figure 7—Fracture Geometry. Application of HVFR in the field. Scenario 1.

Table 4—Parameters of Fracture Geometry. Scenario 1.

Parameter	Value	Dimension
Propped Frac Length	46	feet
Propped Frac Height	131	feet
Propped Width (Well) - Avg.	0.30	Inches
Propped Width (Pay Zone) - Avg.	0.37	Inches
Areal proppant concentration at closure	3.5	Lbm/feet^2
Fracture Conductivity at closure	6291	milidarcy-feet
Dimensionless fracture conductivity	<1	Dimensionless

Details of Scenario 2

the fracture model was calibrated by the analysis of minifrac test summarized in the table 5. This job was executed in a laminar layer (14 feet perforated) located in the upper sand of reservoir. At 14 bpm, the maximum surface pressure and bottomhole pressure were 2970 psi and 3820 psi, respectively. By the analysis of pressure declination of minifrac, the matching of surface and bottomhole pressures were performed updating the fracturing model through the fracturing Software (Figure 8). The fluid efficiency was 22% and the frictions inside of tubing were 99 psi/1000 feet at 14 bpm. In summary, 270 Bbl of HVFR and 14K pounds of 16/30 natural sand were pumped (Figure 9). A tip screen-out packing of the fracture was evidenced by mode III in the net pressure. The fracture geometry shows a length growth supported by the high contrast in the geomechanical parameters (figure 10). The main parameters of fracture geometry estimated in the scenario 2 are showed in the table 6.

Table 5—Analysis of Minifrac Test. Scenario 2.

Parameter	Value	Dimension
Measured Depth	2752	feet
True Vertical Depth	2743	feet
Surface Pre-ISIP (Instantaneous Shut-In Pressure)	1207	Psi
Surface ISIP	750	Psi
Bottomhole ISIP	1938	Psi
Fracture Gradient	0.71	Psi/feet
Closure Stress	1561	Psi
Closure Stress Gradient	0.57	Psi/feet
Closure Time	8.7	Minutes
Net Pressure	372	Psi
Fluid Efficiency	22	%
Total Friction at 30 bpm	457	Psi
Frac String friction at 30 bpm	275	Psi
Entry Hole friction at 30 bpm	182	Psi

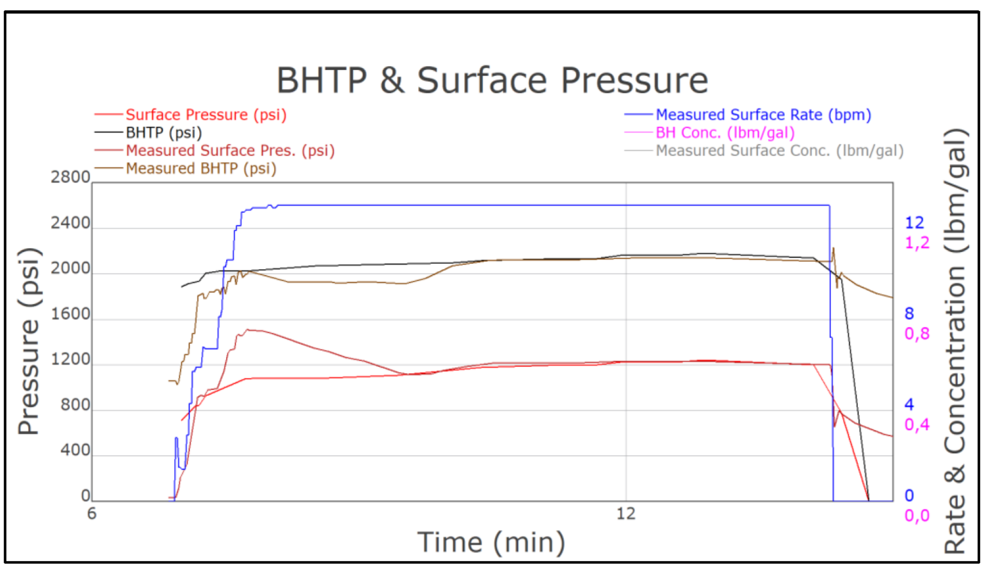

Figure 8—Match of Pressures. Minifrac. Scenario 2.

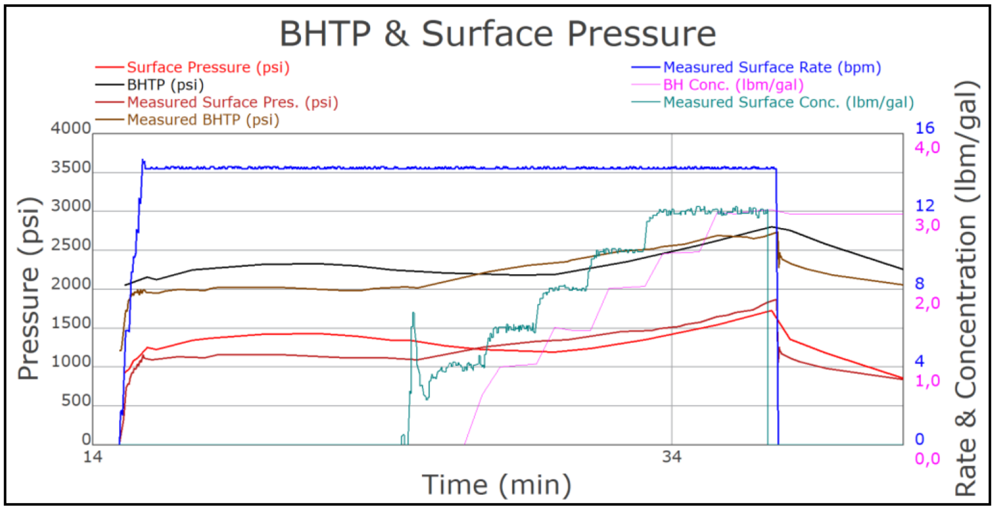

Figure 9—Parameters of Fracturing Job. Application of HVFR in the field. Scenario 2.

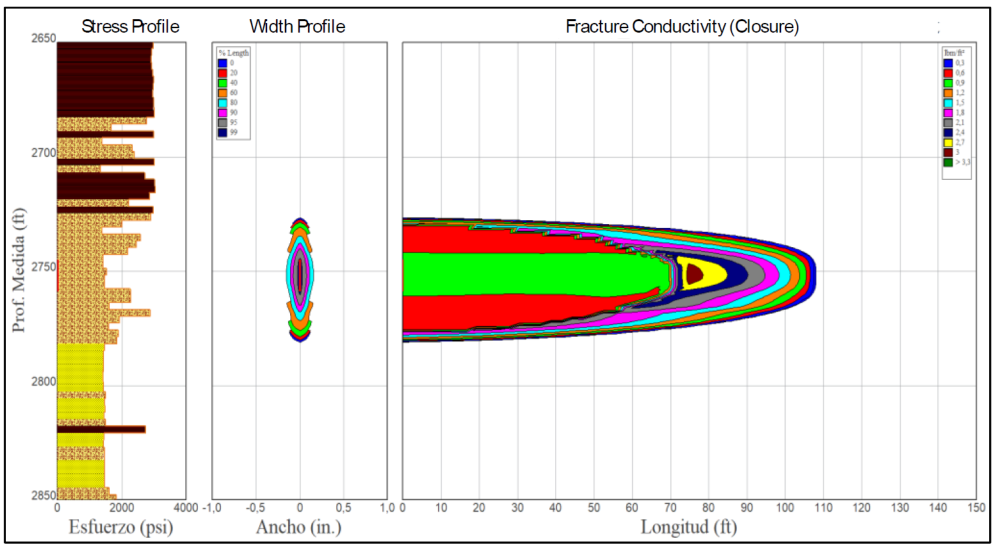

Figure 10—Fracture Geometry. Application of HVFR in the field. Scenario 2.

Table 6—Parameters of Fracture Geometry. Scenario 2.

Parameter	Value	Dimension
Propped Frac Length	107	feet
Propped Frac Height	44	feet
Propped Width (Well) - Avg.	0.11	Inches
Propped Width (Pay Zone) - Avg.	0.16	Inches
Areal proppant concentration at closure	1.3	Lbm/feet^2
Fracture Conductivity at closure	5772	milidarcy-feet
Dimensionless fracture conductivity	1.5	Dimensionless

Conclusions

Formulations of the next generation of HVFR for the scenarios 1 and 2 reviewed along this paper demonstrated optimal hydration with the base fluid (injection and fresh waters respectively), which in turn ensured adequate viscosity (>250 cp) for proppant transport and proppant distribution within the fracture.

The reduction of components (additives) to form the next generation of HVFR system makes it less expensive than conventional crosslinked fluids. This also results in optimization of equipment needed (for example, elimination of the hydration unit).

The proppant carrying ability of the fracturing fluid was found to be adequate during field execution. This allowed obtaining fracture geometry characteristics like that previously established in scenarios 1 and 2.

The fluid breakout was ideal, obtaining viscosities of <10 cP, which allowed a better performance in the fluid flow back times in both reservoirs (scenarios 1 y 2).

The shear stress values were reduced in the pumping BHA, which allowed low friction loss rates (<2300psi) to be obtained for an average flow rate of 35 bpm, which shows a greater pressure impact within the fracture than conventional fluids with Guar Gum, optimizing the HHP on the surface.

Acknowledgments

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Nomenclature

HVFR: High Viscosity Friction Reducer
HHP: Hydraulic Horsepower
Leak-off: Rate of fluid loss to the formation in each period.
Cp: Unit of measurement of viscosity equivalent to one hundredth of a poise, whose symbol is cp.
ppa: Unit of concentration pound of sand added to one gallon of fluid.
gpt: Unit of volume concentration, gallons per thousand gallons of fluid.
mD: millidarcy. Standard unit of permeability measurement.
PAD: Hydraulic Fracturing Stage to get formation breakdown.
WHP: Well Head Pressure.
Keo and Kwo: Effective Permeability to oil and water, respectively.

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